

Loan and contribution delinquency workflow

This document describes how FundFlow tracks late contributions, overdue loan installments, member delinquency, and guarantor liability transfer.

What was already there

Area	Backend
Late fees (tiered by days)	LateFeeService, contribution/loan settings
Guarantor debits after grace	LoanDefaultService (daily <code>loans:check-defaults</code>)
Late contributions on apply	ContributionService (<code>is_late</code> , <code>late_fee_amount</code>)
Member delinquent status	Model + portal block — but not auto-synced
Notifications	LoanDefaultWarningNotification, LoanDefaultGuarantorNotification

Key files (pre-existing)

- `app/Services/Loans/LateFeeService.php`
- `app/Services/Loans/LoanDefaultService.php`
- `app/Services/Loans/LoanRepaymentService.php`
- `app/Services/ContributionService.php`
- `app/Console/Commands/LoansCheckDefaultsCommand.php`
- `routes/console.php` — `loans:check-defaults` daily at 07:00

What was missing (now built)

Previously, most logic lived in services and cron but was **not wired end-to-end** — notably, installments were never set to **overdue**, and `guarantor_liability_transferred_at` was never set from the UI.

1. LoanDelinquencyService

Path: `app/Services/Loans/LoanDelinquencyService.php`

Method	Purpose
<code>markOverdueInstallments()</code>	Pending installments past the contribution-cycle deadline → overdue + late fee
<code>syncMemberDelinquencyStatus()</code>	active delinquent based on overdue installments or unpaid posted contributions
<code>transferGuarantorLiability()</code>	Sets <code>guarantor_liability_transferred_at</code> (immediate guarantor collection path)
<code>restoreBorrowerLiability()</code>	Clears <code>guarantor_liability_transferred_at</code> (standard warning cycle)
<code>runDailyMaintenance()</code>	Full pipeline used by <code>loans:check-defaults</code>
<code>unpaidContributionPeriods()</code>	Per-member list of overdue contribution periods (respects <code>joined_at</code>)
<code>contributionArrearsTableRecords()</code>	Flat rows for the Contribution arrears Filament table
<code>filterContributionArrearsRecords()</code>	Search/sort for array-based table records
<code>memberArrearsSummary()</code>	Snapshot for member view + member dashboard banner
<code>digestCounts()</code>	KPIs for insights widget and admin digest
<code>syncMemberDelinquencyStatusForMember()</code>	Single-member delinquency sync from admin UI

`loans:check-defaults` now runs: **mark overdue** → **sync members** → **warnings** / **guarantor debits**.

2. Admin UI (distributed by domain)

Delinquency is no longer a standalone page. Use:

Concern	Where	Tab / action
Overdue loan installments	Loans list	Overdue installments
Guarantor exposure	Loans list	Guarantor exposure
Contribution arrears	Contributions list	Arrears
Delinquent members	Members list	Delinquent
Member status (sync / restore)	Member edit	Delinquency action group
Policy thresholds	Settings → Contributions	Delinquency policy section

Shared tables: `app/Filament/Support/LoanDelinquencyTables.php`

Maintenance actions: `app/Filament/Support/LoanDelinquencyHeaderActions.php` on **Loans** (and **Contributions** → **Arrears** tab)

- **Run delinquency check** — `full runDailyMaintenance()`
- **Mark overdue only** — `markOverdueInstallments()` only
- **Send admin digest** — notifies tenant admins when there is activity to review

Insights: `LoanInsightsService::delinquencySnapshot()` on loan list tabs; KPI links route to the tabs above.

3. Loan view actions

File: `app/Filament/Support/LoanFilamentActions.php`

- **Transfer liability to guarantor** (requires overdue installments + assigned guarantor)
- **Restore borrower liability**

View loan summary (`ViewLoan.php`): guarantor liability date, late repayment count.

4. Tests

File: `tests/Feature/Tenant/LoanDelinquencyServiceTest.php`

Covers: mark overdue, member delinquent sync, guarantor liability transfer, unpaid period labels, periods before `joined_at` excluded, per-period table rows with `pending` status.

5. Localization

Arabic strings for main UI labels in `lang/ar.json` (e.g. Delinquency, Overdue installments, Transfer liability to guarantor).

How the workflow runs

Borrower path

For each overdue installment on active loans:

1. Count defaults against `late_repayment_count` and `default_grace_cycles` (default **2**, from loan settings).
2. While within grace → **warn borrower** (`LoanDefaultWarningNotification`).
3. After grace → **debit guarantor fund** and mark installment paid by guarantor (`LoanDefaultGuarantorNotification`).

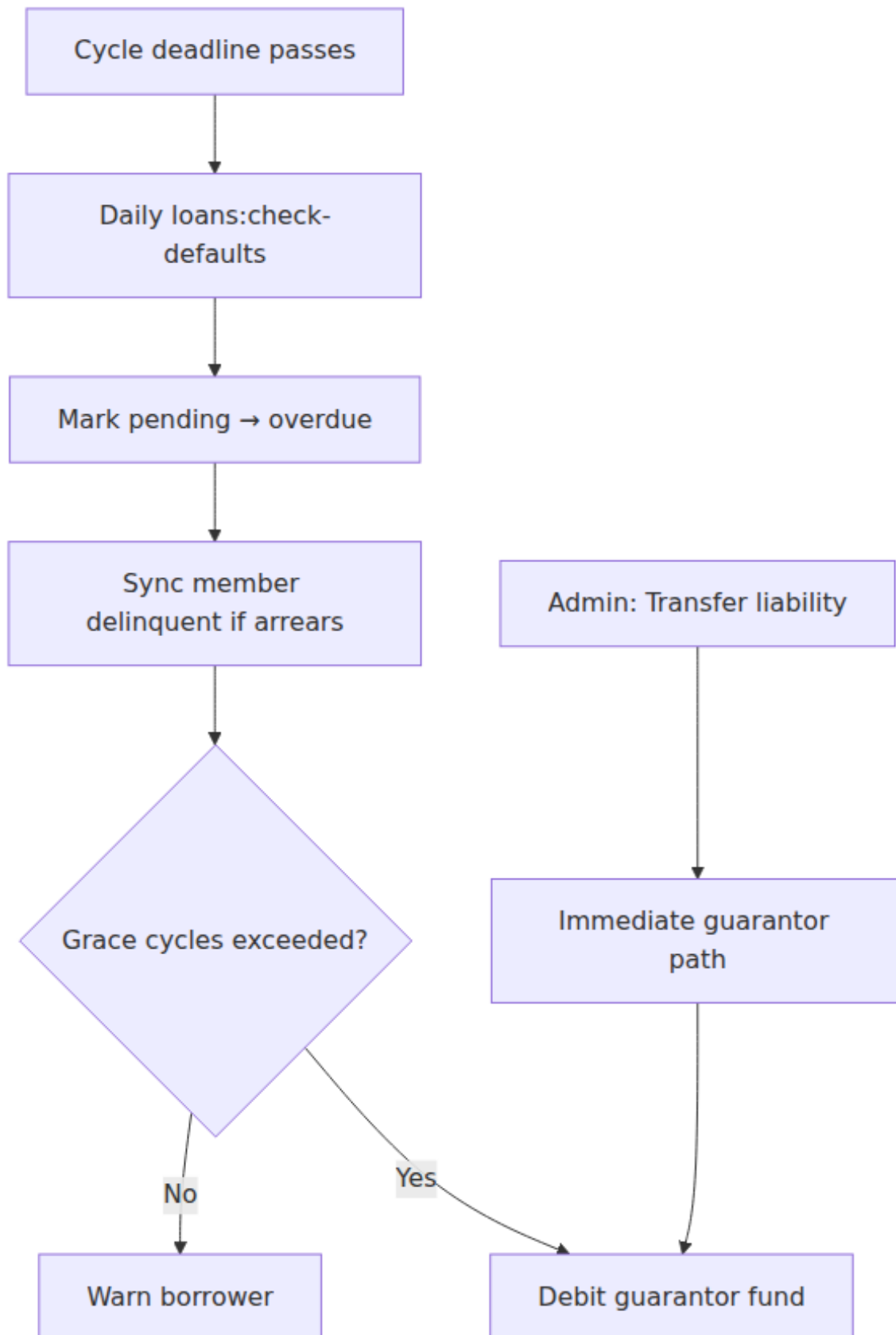


Figure 1: Diagram 1

Admin transfer path

When an admin uses **Transfer liability to guarantor**:

- Sets `guarantor_liability_transferred_at`.
- `LoanDefaultService` uses the **immediate guarantor collection** branch for future overdue installments (skips borrower warning cycle).

Restore borrower liability clears the timestamp and returns to the standard cycle.

Contributions

- Late fees apply when a contribution is applied after the cycle deadline (`ContributionService`).
- **Contribution arrears** tab lists members without a **posted** contribution for any period whose deadline has passed (24-month lookback).
- Members with arrears or overdue installments may be auto-marked **delinquent** (portal blocked); status restores to **active** when arrears clear.

Member delinquency

`syncMemberDelinquencyStatus()` marks members **delinquent** when:

- They have **overdue** installments on any active loan, or
- They have **contribution arrears** (missing posted contribution after deadline).

Restores to **active** when neither condition applies.

Scheduled jobs (contributions & loans)

From `routes/console.php`:

Command	Schedule
<code>contributions:notify</code>	Monthly, 1st, 09:00
<code>contributions:apply</code>	Monthly, 5th, 09:00
<code>loans:send-due-notifications</code>	Monthly, 1st, 08:00
<code>loans:apply-repayments</code>	Monthly, 6th, 06:00
<code>loans:check-defaults</code>	Daily , 07:00
<code>delinquency:send-digest</code>	Daily , 07:30 (after defaults check)

Related model fields

Loan

- `late_repayment_count`
- `guarantor_liability_transferred_at`
- `guarantor_member_id`

LoanInstallment

- `status`: pending | overdue | paid
- `is_late`, `late_fee_amount`
- `paid_by_guarantor`

Contribution

- `is_late`, `late_fee_amount`
- `status`: must be posted to clear arrears

Member

- **status:** includes **delinquent** (portal blocked with suspended/withdrawn/terminated)

Follow-up summary

This section records work completed after the initial delinquency workspace shipped — including fixes for mis-aligned Contribution arrears data and admin/member UX polish.

1. Contribution arrears alignment (fixed)

Problem: The Contributions tab listed one row per member with many periods crammed into a single cell (`HtmlString`), which broke table layout and showed periods before a member's `joined_at`.

Solution:

Change	Detail
Row model	One Filament row per member + period via <code>contributionArrearsTableRecords()</code> and <code>->records()</code>
Eligibility	<code>memberLiableForContributionPeriod()</code> skips months before <code>joined_at</code>
Posted only	Only posted contributions clear arrears; UI shows missing, pending, or failed
Columns	Member, Period, Contribution (badge), Monthly, Late fee, Member status (optional)
Search/sort	<code>filterContributionArrearsRecords()</code> ; default sort by year/month descending
Tab switching	<code>wire:key="delinquency-table-{tab}" + getTableQueryStringIdentifier()</code>

Key files: `LoanDelinquencyService.php`, `LoanDelinquencyTables.php` (`configureContributionArrearsTable()`), `ListContributions.php` (Arrears tab)

2. Member admin actions

On **Members** → **View** (`MemberDelinquencyActions` + infolist section):

Action	Behavior
Sync delinquency status	<code>syncMemberDelinquencyStatusForMember()</code> — promote to delinquent or restore when arrears clear
Mark delinquent	Manual delinquent flag (blocks member portal)
Restore active	Restore when arrears cleared; optional force restore
Delinquency workspace	Opens Loans → Delinquency

Delinquency infolist block: overdue installment count, unpaid contribution period labels.

Key files: `app/Filament/Support/MemberDelinquencyActions.php`, `EditMember.php`

3. Admin delinquency digest

Piece	Detail
Notification	<code>DelinquencyDigestNotification</code> (database + mail when admin email is set)
Recipients	All tenant users with <code>is_admin = true</code>
Content	Counts of overdue installments, contribution arrears periods, delinquent members; link to delinquency workspace
Command	<code>php artisan delinquency:send-digest</code>
Schedule	Daily 07:30 (after <code>loans:check-defaults</code> at 07:00)
Manual trigger	Send admin digest header action on delinquency page

Key files: `DelinquencyDigestService.php`, `DelinquencyDigestNotification.php`, `DelinquencySendDigestCommand.php`, `routes/console.php`

4. Member panel arrears banner

Piece	Detail
Widget	<code>MemberArrearsAlert</code> on member dashboard
Visibility	Shown when <code>memberArrearsSummary()['has_arrears']</code> is true
Copy	Overdue installment count; unpaid contribution periods; delinquent messaging
Links	My loans, My contributions

Key files: `app/Filament/Member/Widgets/MemberArrearsAlert.php`, `resources/views/filament/member/widgets/mem`

5. Verification

`php artisan test --compact tests/Feature/Tenant/LoanDelinquencyServiceTest.php`

Six tests cover overdue marking, delinquent sync, guarantor liability transfer, period labels, `joined_at` exclusion, and per-period table rows with contribution status.

6. Localization

Additional Arabic keys in `lang/ar.json` for delinquency workspace, digest, member actions, and member banner strings.

Possible future enhancements

- Email channel for delinquency digest (in addition to in-app/database notifications)
- Per-member filter on the delinquency contributions tab
- Insights polling (only if reintroduced with stable route-based `context`, not a fixed default)